

Answer **all** questions.

1. (a) Rock salt is a mixture of salt and sand. Crystals of pure salt can be obtained from rock salt.

A–E are the steps in the method used but they are in the wrong order.

- A** Add water to a sample of rock salt in a beaker and stir
- B** Heat the solution to evaporate some of the water
- C** Grind the rock salt into a fine powder
- D** Filter the mixture to separate the sand from the salt solution
- E** Leave the saturated solution in a warm place for a few days so that crystals of salt form

Put the steps in the correct order. The first step is already included.

[2]

C

first step

- (b) A student was asked to investigate the dyes present in an orange sweet.

The student carried out the following method. There are two errors in the method.

- Draw a line using a ruler and pen on chromatography paper.
- Place a sample of the orange colour on the line.
- Stand the chromatography paper in a beaker and add enough water to just cover the sample.
- Leave the paper to stand until the water rises to the top of the paper.

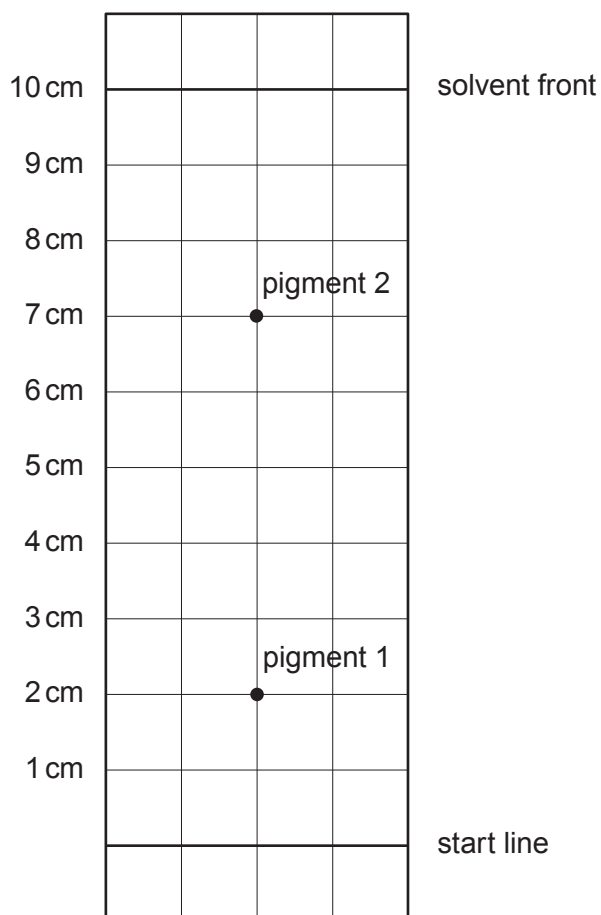
- (i) State the **two** errors in the method.

[2]

1.
2.



(ii) Another student used a correct method and obtained the chromatogram below.



Use the formula to calculate the R_f value for pigment 2.

[2]

$$R_f = \frac{\text{distance travelled by the pigment}}{\text{distance travelled by the solvent front}}$$

$R_f = \dots\dots\dots$

